

A Counterexample to the Proposed Stability Conditions for Gaussian Lattice Sums

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11 August 2026

Status. Candidate full counterexample to the Appendix C conjectured sufficient conditions in arXiv:2110.06008, likely valid, for human review.

1 Source conjecture

For $\tau = x + iy$ in

$$D_+ = \{(x, y) : 0 \leq x \leq \tfrac{1}{2}, x^2 + y^2 \geq 1, y > 0\},$$

Bétermin, Faulhuber, and Steinerberger parametrize a unit-area lattice and define

$$\theta_{x,y}(z; \alpha) = \sum_{k,l \in \mathbb{Z}} \exp \left[-\frac{\pi \alpha}{y} \left((k + z_1)^2 + 2x(k + z_1)(l + z_2) + (x^2 + y^2)(l + z_2)^2 \right) \right]. \quad (1)$$

The hexagonal lattice has $(x, y) = (1/2, \sqrt{3}/2)$ and deep hole $z^- = (1/3, 1/3)$. Appendix C proposes that

$$\theta_{1/2, \sqrt{3}/2}(z^-; \alpha) \geq \theta_{x,y}(\tilde{z}(x, y); \alpha) \quad (\alpha > 0) \quad (2)$$

at least when the selector obeys

$$\tilde{z}_1 + x\tilde{z}_2 = \tfrac{1}{2}, \quad \partial_x \tilde{z}_2 \leq 0, \quad \tilde{z}(\tfrac{1}{2}, y) = a(\tfrac{1}{2}, y), \quad \tilde{z}(0, y) \in [\tfrac{1}{4}, \tfrac{1}{2}], \quad (3)$$

and tends to z^- with the lattice [1, Appendix C].

C. A CONJECTURE ON THE STABILITY OF $z_{\Lambda_2}^-$

Our family of theta functions shows – at least numerically, but also in our proofs – some stability against controlled perturbation, a property which is provably false for the theta functions studied by Montgomery [61]. We observed that the following inequality seems to hold:

$$(C.1) \quad \theta_{\Lambda_2}(z_{\Lambda_2}^-; \alpha) \geq \theta_{\Lambda}(\tilde{z}; \alpha), \quad \forall \alpha > 0, \quad \text{and} \quad \tilde{z} \rightarrow z_{\Lambda_2}^- \text{ as } \Lambda \rightarrow \Lambda_2.$$

We are not able to give general conditions on how $\tilde{z}$ must approach $z_{\Lambda_2}^-$ as Λ tends to the hexagonal lattice Λ_2 in the space of lattices, but our proofs make us believe that (C.1) holds at least under the following conditions;

$$\tilde{z}_1 + x \tilde{z}_2 = \frac{1}{2}, \quad \partial_x \tilde{z}_2 \leq 0, \quad \tilde{z}(\tfrac{1}{2}, y) = a(\tfrac{1}{2}, y) \quad \text{and} \quad \tilde{z}(0, y) \in [\tfrac{1}{4}, \tfrac{1}{2}].$$

This is under the hypothesis that $(x, y) \in D_+$. We remark that the circumcenter a meets the above criteria, as well as our point b . The difference between the points a and b is that $\partial_x b_2 \equiv 0$ whereas $\partial_x a_2 < 0$ for $x \in (0, \frac{1}{2})$. We note that $\partial_x a_2|_{x=\frac{1}{2}} = \partial_x b_2|_{x=\frac{1}{2}} = 0$, but we have no indication that this property is needed for the perturbation result.

The last membership in (3) is dimensionally ambiguous for a vector-valued selector; the construction below has $\tilde{z}(0, y) = (1/2, 1/2)$, so it satisfies both the natural reading as a condition on $\tilde{z}_2$ and the coordinatewise reading.

2 Claimed counterexample

Theorem 1. *Define on all of D_+*

$$\tilde{z}_2(x, y) = \frac{1}{2} - \frac{[4x(1-x)]^{100}}{8y^2}, \quad \tilde{z}_1(x, y) = \frac{1}{2} - x\tilde{z}_2(x, y). \quad (4)$$

This selector satisfies every condition in (3), tends to $(1/3, 1/3)$ at the hexagonal lattice, and also satisfies the stronger boundary condition

$$\partial_x \tilde{z}_2(\tfrac{1}{2}, y) = 0.$$

Nevertheless, at

$$x = \frac{9}{20}, \quad y = \frac{\sqrt{319}}{20}, \quad \alpha = 1, \quad (5)$$

one has the strict reverse inequality

$$\theta_{x,y}(\tilde{z}(x, y); 1) > \theta_{1/2, \sqrt{3}/2}((1/3, 1/3); 1).$$

Hence the proposed sufficient conditions do not imply (2), even after adding the boundary-criticality property discussed immediately after the conjecture.

3 Idea of the proof

The endpoint constraints control how far the selector can move but do not control how rapidly it moves as x leaves the hexagonal boundary. The factor $[4x(1-x)]^{100}$ equals one at $x = 1/2$, has

zero derivative there, yet drops quickly just inside the domain. Thus (4) remains perfectly smooth and monotone while moving much closer to $(1/2, 1/2)$ than the canonical selectors considered in the source paper. That displacement raises the theta value enough to cross the hexagonal benchmark.

To make the crossing rigorous, no infinite numerical sum is trusted. We keep only a 5×5 square of positive terms for the candidate, giving a lower bound. For the hexagonal sum, its quadratic form dominates a separable one; the omitted complement is then bounded by two elementary one-dimensional Gaussian geometric tails. Directed interval arithmetic leaves a gap above 0.003847.

4 Proof

Put $G(x) = 4x(1 - x)$. On $0 \leq x \leq 1/2$ one has $0 \leq G \leq 1$ and $G'(x) = 4(1 - 2x) \geq 0$. The line condition in (3) is built into (4), while

$$\partial_x \tilde{z}_2 = -\frac{100}{8y^2} G(x)^{99} G'(x) \leq 0.$$

At $x = 0$, $G = 0$ and $\tilde{z} = (1/2, 1/2)$. At $x = 1/2$, $G = 1$ and

$$\tilde{z}_2 = \frac{1}{2} - \frac{1}{8y^2}, \quad \tilde{z}_1 = \frac{1}{4} + \frac{1}{16y^2},$$

which are exactly the source paper's circumcenter coordinates $a(1/2, y)$. Moreover $G'(1/2) = 0$, proving boundary criticality. Continuity at $(1/2, \sqrt{3}/2)$ gives $\tilde{z} \rightarrow (1/3, 1/3)$. Thus all selector claims are proved.

For the test point (5), $x^2 + y^2 = 1$, so it lies in D_+ , and

$$G(x) = \frac{99}{100}, \quad \tilde{z}_2 = \frac{1}{2} - \frac{50}{319} \left(\frac{99}{100} \right)^{100}, \quad \tilde{z}_1 = \frac{1}{2} - \frac{9}{20} \tilde{z}_2. \quad (6)$$

Let C_2 be the sum in (1) at (5) and (6), restricted to $-2 \leq k, l \leq 2$. Positivity gives

$$\theta_{x,y}(\tilde{z}; 1) \geq C_2. \quad (7)$$

It remains to give a rigorous upper bound for the hexagonal benchmark. Write $u = k + 1/3$ and $v = l + 1/3$. Its exponent uses

$$q_H(u, v) = \frac{2}{\sqrt{3}}(u^2 + uv + v^2) \geq \frac{u^2 + v^2}{\sqrt{3}}.$$

Set $c = \pi/\sqrt{3}$ and, for $N \geq 0$,

$$A_N = \sum_{|n| \leq N} e^{-c(n+1/3)^2},$$

$$T_N = \frac{2e^{-c(N+2/3)^2}}{1 - e^{-c(2N+7/3)}}.$$

Lemma 1. *The complement of the square $|k|, |l| \leq N$ in the hexagonal theta sum is at most $2(A_N + T_N)T_N$.*

Proof. The one-dimensional tail satisfies

$$\sum_{|n|>N} e^{-c(n+1/3)^2} \leq 2 \sum_{n=N+1}^{\infty} e^{-c(n-1/3)^2} \leq T_N.$$

Indeed, the first dominating term is $e^{-c(N+2/3)^2}$, and the ratio of successive terms is at most $e^{-c(2N+7/3)}$. The full one-dimensional sum is consequently at most $A_N + T_N$. By $q_H \geq (u^2 + v^2)/\sqrt{3}$ and a union bound over the two coordinate tails, the two-dimensional complement is at most $2(A_N + T_N)T_N$. $\square$

Let H_2 be the 5×5 hexagonal square. The directed 60-decimal interval calculation in the accompanying checker (file `code/check_counterexample.py`) gives

$$\begin{aligned} C_2 &> 0.9242317405007238239445, \\ H_2 &< 0.9203713649781851694427, \\ 2(A_2 + T_2)T_2 &< 0.000013118172843186372. \end{aligned}$$

Every quantity is a finite composition of rational arithmetic, square roots, π , and exponentials, evaluated with outward rounding. Therefore

$$\begin{aligned} \theta_{x,y}(\tilde{z}; 1) - \theta_{1/2, \sqrt{3}/2}((1/3, 1/3); 1) &\geq C_2 - H_2 - 2(A_2 + T_2)T_2 \\ &> 0.00384725734969546 > 0. \end{aligned}$$

Together with (7) and Lemma 1, this proves the strict reversal and completes the proof of Theorem 1.

Corollary 1. *For the selector (4), the conjectured inequality fails on an open neighborhood of the parameter triple in (5).*

Proof. The theta series converges locally uniformly in (x, y, z, α) for $y > 0$. The strict gap just proved therefore persists by continuity. $\square$

5 Verification, scope, and novelty

The interval checker is deterministic and asserts positivity of the lower endpoint of the certified gap. The separate `verification.md` audits the selector conditions, tail estimate, and possible interpretations of the source’s final boundary condition.

This counterexample refutes only the proposed sufficient conditions in Appendix C. It does not affect the paper’s proved variational theorem, the Strohmer–Beaver application, or the fact that the hexagonal deep hole is a universal cold spot. A corrected stability statement would need a quantitative modulus controlling how fast $\tilde{z}$ may leave the circumcenter as the lattice leaves the hexagonal boundary.

A bounded novelty search on 11 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; exact-title and arXiv-id searches; and web searches combining “stability conjecture”, “widetilde z”, and “counterexample”. Later work on stable cold spots cites the source

theorem and emphasizes that exact cold spots are difficult to determine [2], but the search found no resolution of this Appendix selector conjecture. This supports, but cannot certify, novelty.

Human-review recommendation. Review as a full counterexample. The only computational component is the explicit finite interval inequality; its analytic tail enclosure is included above and can be independently reproduced in under a second.

References

- [1] L. Bétermin, M. Faulhuber, and S. Steinerberger, *A variational principle for Gaussian lattice sums*, arXiv:2110.06008 (2021), Appendix C, <https://arxiv.org/abs/2110.06008>.
- [2] E. Bannai, L. Bétermin, and F. Vallentin, *Polarization of lattices: Stable cold spots and spherical designs*, Forum of Mathematics, Sigma (2026).